MONTH 1 LEARNING PLAN

UI, Modern CSS & React Foundations

A visual study guide for building, reviewing, and debugging with confidence.

Build

Semantic HTML, responsive layouts, data demos, and a React task board.

Understand

Box model, Flexbox, immutable data, asynchronous requests, state, and data flow.

Validate

Use DevTools, precise error reports, code review checklists, and independent diagnosis.

> *Your job with AI-assisted software is to review, validate, and debug what was generated.*

Source: Month 1 Self-Study Guide: UI, Modern CSS & React Foundations

AT A GLANCE

Four-Week Roadmap

WEEK 1

Modern Layouts

Master the box model, Flexbox axes, responsive constraints, and DevTools inspection.

01 Box model and `border-box`

02 Flexbox alignment

03 Responsive widths

04 Inspect winning CSS rules

05 Build a two-column page

WEEK 2

Data Flow

Transform data without mutation and handle asynchronous requests visibly.

01 Values and references

02 `map`, `filter`, and spread

03 `fetch` lifecycle

04 Network debugging

05 Build a data demo

WEEK 3

React Architecture

Build predictable components with props, state, keys, and client boundaries.

01 Component hierarchy

02 State and events

03 Immutable collections

04 Derived state and keys

05 Build a task board

WEEK 4

Review & Integration

Prompt precisely, diagnose independently, review AI output, and teach back the system.

01 Prompt architecture

02 Code review drill

03 Debugging drill

04 Refactor components

05 Capstone and teach-back

Daily Session Pattern

10 min Review → **15 min** Concept → **25 min** Practice → **10 min** Document

One small sandbox project should grow throughout the month.

MENTAL MODELS

How the Core Ideas Fit Together

HTML structure

html
root

↓

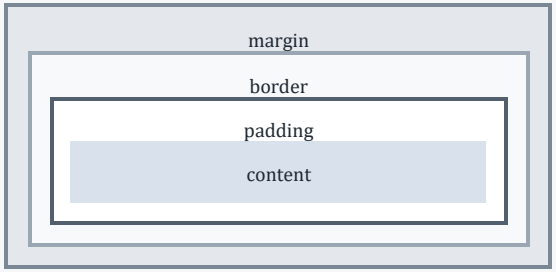
head
metadata and links

↓

body
visible content

```
<header> <main> <footer>
```

CSS box model



border-box includes padding and borders inside the declared size.

```
* { box-sizing: border-box; }
```

Flexbox axes

- **justify-content** follows the main axis.
- **align-items** follows the cross axis.
- **flex-direction** chooses the main axis.

JavaScript transformation

React data flow


```
setCount(prev => prev + 1)
```

Immutable update

References matter. Create a new outer array or object so React can detect the change.

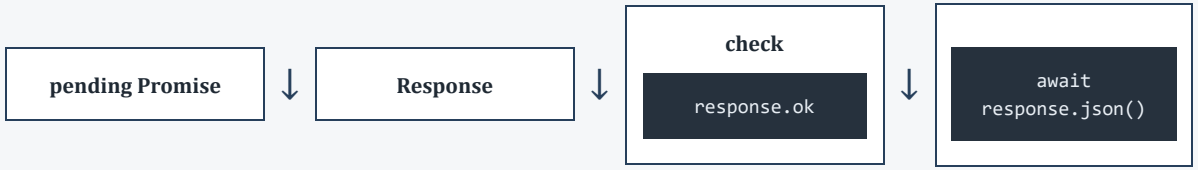
```
setItems([...items, newItem])

items.map(update)
items.filter(keep)
```

Remember: derived values should be calculated from existing state instead of stored redundantly.

DEBUGGING WORKFLOW

Trace the Request, Then Explain the Failure

Fetch lifecycle



Important: HTTP 400 and 500 responses usually do not reject

```
fetch
```

; check

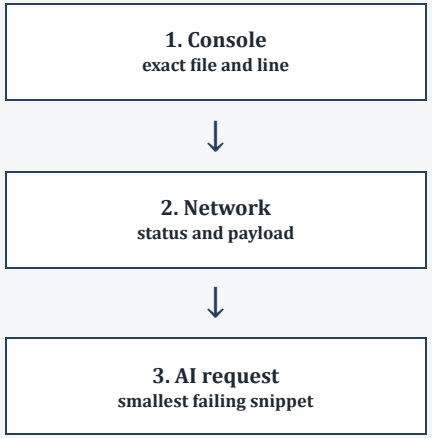
```
response.ok
```

yourself.

Network inspection

- **Headers:** URL, method, and status.
- **Payload:** JSON sent by the client.
- **Response:** raw JSON returned by the server.
- **States:** loading, success, empty, and error.

Systematic error triage



AI review checklist

[ ]

```
use client
```

when hooks or events are used.

[ ] State updates are immutable.

[ ] List keys are stable and unique.

[ ] Layouts use responsive constraints.

[ ] Derived values are not redundant state.

> *Make one independent diagnosis before requesting generated code.*

Record the exact error, expected behavior, and smallest relevant snippet.

COMPLETION AND RETENTION

Evidence, Review, and Month-End Check

Daily evidence

[ ] One code sample.

[ ] One screenshot or console/network observation.

[ ] One sentence explaining the lesson.

[ ] One independent diagnosis before asking AI.

Spaced review

Day 1 first review → **Day 3** second review → **Day 7** retest

Mark a term learned only after defining and applying it correctly twice without notes.

Portfolio evidence

[ ] Week 1 responsive layout.

[ ] Week 2 data demo.

[ ] Week 3 React task board.

[ ] Week 4 reviewed capstone.

Month 1 Self-Assessment

[ ] Explain why

```
border-box
```

matters.

[ ] Build a responsive two-column layout.

[ ] Explain mutation versus shallow copy.

[ ] Explain why

```
push()
```

can prevent a React update.

[ ] Diagnose a failed network request.

[ ] Resolve a missing

```
use client
```

boundary.

Before Month 2, answer every item “Yes” or create a specific remediation task.